

Group Coursework Submission Form

Specialist Masters Programme

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MSc in: Actuarial Management/Actuarial Science	
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Lecturer: Dr. Iqbal Owadally	Submission Date: 28/02/2025
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Deduction for Late Submission of assignment:

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For Students:
Once marked please refer to Moodle
for your final coursework grade,
including your Peer Assessment grade.

CM2 Coursework

Group 10

February 28, 2025

Question 1

The selected investment portfolio is constructed from the stocks of five major U.S. technology companies. The historical stock price data was sourced from Investing.com.

The relevant section of the website for equities can be accessed at:
<https://uk.investing.com/equities>

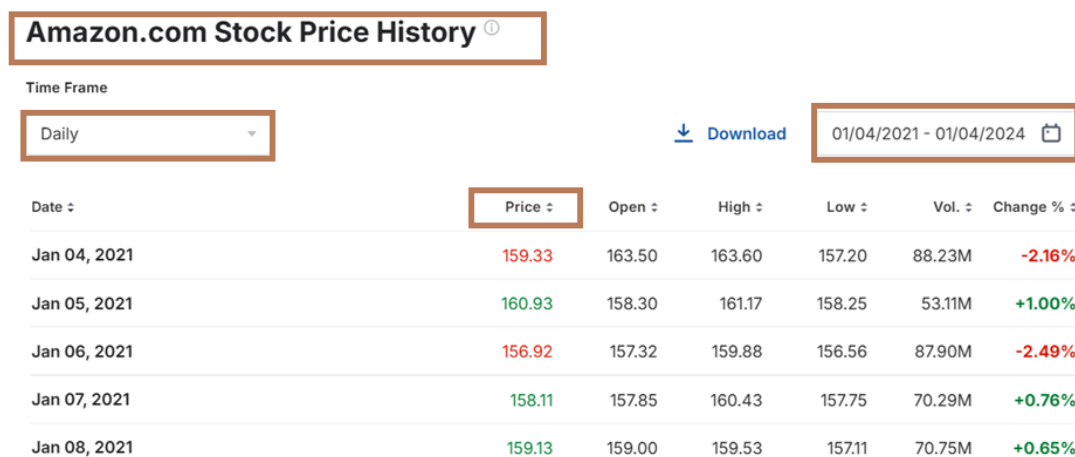
Below is the list of stocks along with direct links to the sources from which the historical data was downloaded:

- Apple Inc (**AAPL**) <https://uk.investing.com/equities/apple-computer-inc>
- Meta Platforms Inc (**META**) <https://uk.investing.com/equities/facebook-inc>
- NVIDIA Corporation (**NVDA**) <https://uk.investing.com/equities/nvidia-corp>
- Microsoft Corporation (**MSFT**) <https://uk.investing.com/equities/microsoft-corp>
- Amazon.com Inc (**AMZN**) <https://uk.investing.com/equities/amazon-com-inc>

The data retrieved spans a period of three years, **from January 4, 2021, to January 4, 2024**. The time frame selected was daily prices, ensuring a comprehensive dataset for analysis.

The price label chosen for this report is the **closing price**. The closing price represents the final price at which a stock is traded on a given trading day. This metric is widely used in financial analysis as it reflects investor sentiment and market valuation at the end of the trading session.

Figure 1 below illustrates the appearance of the historical data. The screenshot is taken from the Investing.com website and specifically shows data for AMZN stock:



Date ↕	Price ↕	Open ↕	High ↕	Low ↕	Vol. ↕	Change % ↕
Jan 04, 2021	159.33	163.50	163.60	157.20	88.23M	-2.16%
Jan 05, 2021	160.93	158.30	161.17	158.25	53.11M	+1.00%
Jan 06, 2021	156.92	157.32	159.88	156.56	87.90M	-2.49%
Jan 07, 2021	158.11	157.85	160.43	157.75	70.29M	+0.76%
Jan 08, 2021	159.13	159.00	159.53	157.11	70.75M	+0.65%

Figure 1: Historic data for AMZN from Investing.com

The daily prices are available in the Excel file called "**SMM068 _ Coursework _ Group10.xlsm**" under the **Q1** worksheet."

Question 2

(a) Mean-variance portfolio optimization is the foundation of Modern Portfolio Theory (MPT). It represents mathematical framework for constructing an investment portfolio that aims to maximize expected return for a given level of risk or minimize risk for a given level of expected return.

There are several key statistics that need to be calculated for it, namely:

- Daily log returns for each of the five US stocks (Apple, Meta, Nvidia, Microsoft, Amazon) using the formula:

$$\text{Log Return} = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (1)$$

where P_t and P_{t-1} are the prices on consecutive days

Typically, daily log-returns are used as an estimate of expected returns because they are time-consistent and additive over different periods.

- The mean (average return) for each stock is calculated below

Stock	Expected Return
AAPL	0.00045103
META	0.00033799
NVDA	0.00171897
MSFT	0.00069516
AMZN	-0.00012876

Table 1: Expected Returns for selected stocks

- Standard deviation (risk) for each stock, which basically represent uncertainty

Stock	Standard Deviation
AAPL	0.017519
META	0.030221
NVDA	0.032974
MSFT	0.017504
AMZN	0.023551

Table 2: Standard Deviations (Risk) for selected stocks

- Covariance matrix and correlation indicate how different stocks move relative to each other, helping in diversification and needed for optimization:

	AAPL	META	NVDA	MSFT	AMZN
AAPL	0.000306919	0.000287055	0.000366546	0.000221577	0.000251134
META	0.000287055	0.000913336	0.000513191	0.000303795	0.000415910
NVDA	0.000366546	0.000513191	0.001087301	0.000393018	0.000462072
MSFT	0.000221577	0.000303795	0.000393018	0.000306379	0.000276461
AMZN	0.000251134	0.000415910	0.000462072	0.000276461	0.000554664

Table 3: Covariance Matrix for selected stocks

	AAPL	META	NVDA	MSFT	AMZN
AAPL	1				
META	0.5421735	1			
NVDA	0.6345139	0.5149784	1		
MSFT	0.7225747	0.5742964	0.6809386	1	
AMZN	0.6086646	0.5843445	0.5950037	0.6706394	1

Table 4: Correlation Matrix for selected stocks

The calculations above are also done in the Excel model, under the worksheet "**Q2a**".

(b) The efficient frontier is a curve representing the set of optimal portfolios that offer the highest expected return for a given level of risk (standard deviation of returns). It is constructed by solving a portfolio optimization problem, which finds the minimum variance portfolio for different levels of expected returns.

The process involves the following steps:

1. Define **target returns**

A set of target returns is required, spanning from the minimum expected return (AMZN = -0.012876%) to the maximum expected return (NVDA = 0.171897%) in the dataset. These target returns are generated through linear approximation, with the step size determined by

$$\text{Step Size} = \left(\frac{\text{Max}_{\text{return}} - \text{Min}_{\text{return}}}{99} \right) \quad (2)$$

2. Solve the **minimum variance portfolio** for each target return

Constraints used in the portfolio optimization are the following:

• **Weights sum to 1:**

$$w_1 + w_2 + \cdots + w_n = 1$$

• **Target return constraint:**

$$w_1 r_1 + w_2 r_2 + \cdots + w_n r_n = R_{\text{target}}$$

• **No short selling:**

$$w_i \geq 0 \quad \forall i$$

3. Implement an optimization function to solve for weights

VBA programming and the solver function are used to determine the optimal weights for the minimum variance portfolio at each target return. In R, the `quadprog::solve.QP()` function is utilized to achieve the same goal, efficiently solving quadratic programming problems to identify the best weight allocation.

4. Compute Portfolio Risk for Each Target Return

Once the optimal weights are found for each target return, we calculate the portfolio risk (standard deviation) using the variance-covariance matrix of asset returns.

- **Portfolio variance formula:**

$$\sigma_p^2 = W^T \Sigma W$$

where:

- W is the vector of portfolio weights

$$W = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

- Σ is the variance-covariance matrix of asset returns.

- **Portfolio standard deviation (risk):**

$$\sigma_p = \sqrt{W^T \Sigma W}$$

The portfolio standard deviation represents the total risk associated with the optimized portfolio.

Figure 2 represents the graph of the efficient frontier, ignoring the portfolios below the global minimum variance portfolio, which represents the point on the efficient frontier where risk is minimized, regardless of the expected return. Any portfolio below this point is considered inefficient because it carries more risk without offering a better return trade-off.

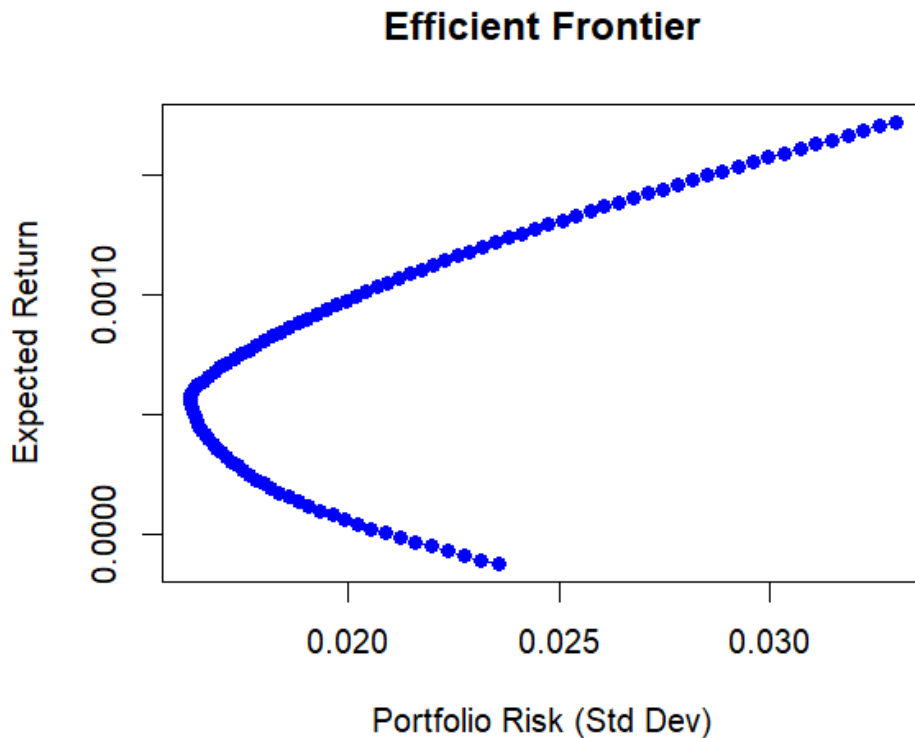


Figure 2: Efficient frontier

In Excel the simulations for the efficient frontier are done on the worksheet "Q2b".

(c) To help the investor build an optimized portfolio using the efficient frontier, the following key considerations should be addressed:

1. **Risk-Return Trade-off & Tolerance**
Higher returns usually come with higher risk. A risk-averse investor may prefer a more stable portfolio with lower expected returns.
2. **Efficient Frontier**
The efficient frontier helps investors choose the best combination of stocks for a given level of risk. Investing outside this range may mean unnecessary risk or lower returns.
3. **Short Selling restriction**
Short selling is restricted, so the frontier curve assumes only positive weights. This means investors can't hedge against losses.
4. **Diversification**
A tech-focused portfolio (e.g., Apple, Microsoft) increases risk. Spreading investments across sectors and mixing uncorrelated assets improves stability and reduces losses.
5. **Market Conditions**
Stock prices are influenced by economic factors like inflation, interest rates, and global events. In inflationary periods, commodities and real estate often perform well.
6. **Long-Term vs. Short-Term Goals**
If the investor is looking for quick gains, they might take higher risks, but if they want steady growth over years, they should focus on stable and strong companies.
7. **Rebalancing Strategy**
Over time, some investments may grow while others shrink, changing the risk level. Rebalancing the portfolio keeps it aligned with the investor's original risk preference.
8. **Tax Implications**
Profits from selling stocks and dividends are taxed. Tax-efficient strategies, like holding investments for longer periods or using tax-advantaged accounts, can help minimize tax.
9. **Investment Costs**
High transaction costs from fees, commissions, and taxes can reduce profits. Using low-cost platforms and trading less helps control expenses.
10. **Liquidity Needs**
Some assets sell more easily than others. Investors needing quick cash should keep part of their portfolio in liquid assets like cash or highly tradable stocks

Question 3

We are given two assets that have a Normally distributed returns as follows:

- Asset 1 return $R_1 \sim N(0.06, 0.0244)$

- Asset 2 return

$$R_2 \sim \begin{cases} N(0, 0.01) & \text{with probability 0.80} \\ N(0.3, 0.01) & \text{with probability 0.20} \end{cases}$$

(a) Asset 2 follows a mixed distribution, specifically a bimodal distribution.

Expected Return

$$E[R_2] = (\mu_1 \times p_1) + (\mu_2 \times p_2) = (0 \times 0.80) + (0.3 \times 0.2) = 6\%$$

Variance

$$\begin{aligned} \text{Var}(R_2) &= p_1\sigma_1^2 + p_2\sigma_2^2 + p_1p_2(\mu_1 - \mu_2)^2 \\ &= (0.8 \times 0.01) + (0.2 \times 0.01) + (0.8 \times 0.2 \times (-0.3)^2) \\ &= 2.44\% \end{aligned}$$

Asset 2 has an expected return of 6% and a variance of 2.4%, identical to Asset 1.

(b) Expected return ($E(R_2)$) and Variance ($\text{Var}(R_2)$) are calculated in R
Below is the R script used to calculate the expected return and variance of Asset 2.

```
1 # means of normal distributions for Asset 2 respectively
2 means = c(0, 0.3)
3 # Probabilities for the 2 distributions for Asset 2
4 prob.2 <- c(0.8, 0.2)
5 # set seed for reproducibility.
6 set.seed(1)
7 # number of simulations to carry out
8 n <- 100000
9 # sample the means of the 2 distribution based on their
  probability
10 sample.mean = sample(means, size = n, replace = TRUE, prob =
  prob.2)
11
12 #Simulating returns for asset 2
13 returns <- rnorm(n, mean = sample.mean, sd = 0.1)
14
15 ER2 <- mean(returns)
16 VarR2 = var(returns)
17
18 cat("Estimated Expected Return (ER2):", ER2, "\n")
19 Estimated Expected Return (ER2): 0.06023716
20 cat("Estimated Variance (VarR2):", VarR2, "\n")
21 Estimated Variance (VarR2): 0.02444028
22
```


(c) The shortfall probability is the probability that the asset return falls below a given benchmark return R_B

Since $R_1 \sim N(0.06, 0.0244)$, it follows a normal distribution with: mean: $\mu_1 = 6\%$, variance: $\sigma_1^2 = 0.0244$ and standard deviation: $\sigma_1 = \sqrt{0.0244} = 0.1562$

Using the standard normal variable transformation:

$$Z = \frac{R_1 - \mu_1}{\sigma_1} \sim N(0, 1)$$

The shortfall probability is then:

$$P(R_1 < R_B) = P\left(Z < \frac{R_B - \mu_1}{\sigma_1}\right)$$

Case (i): Benchmark Return $R_B = 0\%$

$$Z = \frac{0 - 0.06}{0.1562} = \frac{-0.06}{0.1562} = -0.384$$

Using the standard normal cumulative distribution function (CDF) $\Phi(z)$:

$$P(R_1 < 0) = P(Z < -0.384) = \Phi(-0.384)$$

From normal tables:

$$\Phi(-0.384) \approx 0.35$$

So, the **probability of a return lower than 0% is 35%**.

Case (ii): Benchmark Return $R_B = -10\%$

$$Z = \frac{-0.10 - 0.06}{0.1562} = \frac{-0.16}{0.1562} = -1.024$$

From the standard normal table:

$$P(R_1 < -0.10) = P(Z < -1.024) = \Phi(-1.024)$$

$$\Phi(-1.024) \approx 0.15$$

So, the **probability of a return lower than -10% is 15%**.

Final Results in Mathematical Form

$$P(R_1 < 0) = \Phi\left(\frac{0 - 0.06}{0.1562}\right) \approx 0.35$$

$$P(R_1 < -0.10) = \Phi\left(\frac{-0.10 - 0.06}{0.1562}\right) \approx 0.15$$

These probabilities represent the likelihood of Asset 1 underperforming the given benchmarks.

(d) For Asset 2, we have a bimodal distribution, i.e., a mixture of two normal distributions.

Benchmark of 0%:

$$P(X < 0\%) = P\left(Z < \frac{0 - 0}{0.10}\right) = P(Z < 0) = 0.5$$

$$P(X < 0\%) = P\left(Z < \frac{0 - 0.3}{0.1}\right) = P(Z < -3) = 0.0013499$$

$$P(R_2 < 0\%) = 0.80 \times 0.5 + 0.0013499 \times 0.2 = 0.40$$

Benchmark of -10%:

$$P(X < -10\%) = P\left(Z < \frac{-0.1 - 0}{0.10}\right) = P(Z < -1) = 0.15866$$

$$P(X < -10\%) = P\left(Z < \frac{-0.10 - 0.30}{0.1}\right) = P(Z < -4) = 0.00003$$

$$P(R_2 < -10\%) = 0.80 \times 0.15866 + 0.00003 \times 0.2 = 0.1269$$

Asset 2 (bimodal distribution) has a **probability at the 0% benchmark of 40% and at the -10% benchmark of 12.69%**.

Asset 1 has a lower shortfall probability at the 0% benchmark (35.04%) compared to Asset 2 (40.19%). This indicates that Asset 2 is more likely to underperform compared to Asset 1. At the -10% benchmark, Asset 2 has a slightly lower shortfall probability (12.69%) compared to Asset 1 (15.86%), which could suggest that Asset 2 is less likely to experience extreme losses.

(e) The Probability density function for Asset 1 is a smooth curve centred around its mean (6%) whereas the probability density function for asset 2 (bimodal), which peaks around two points ie at 0% and approximately 30% which shows higher uncertainty and variability in Assets 2's return.

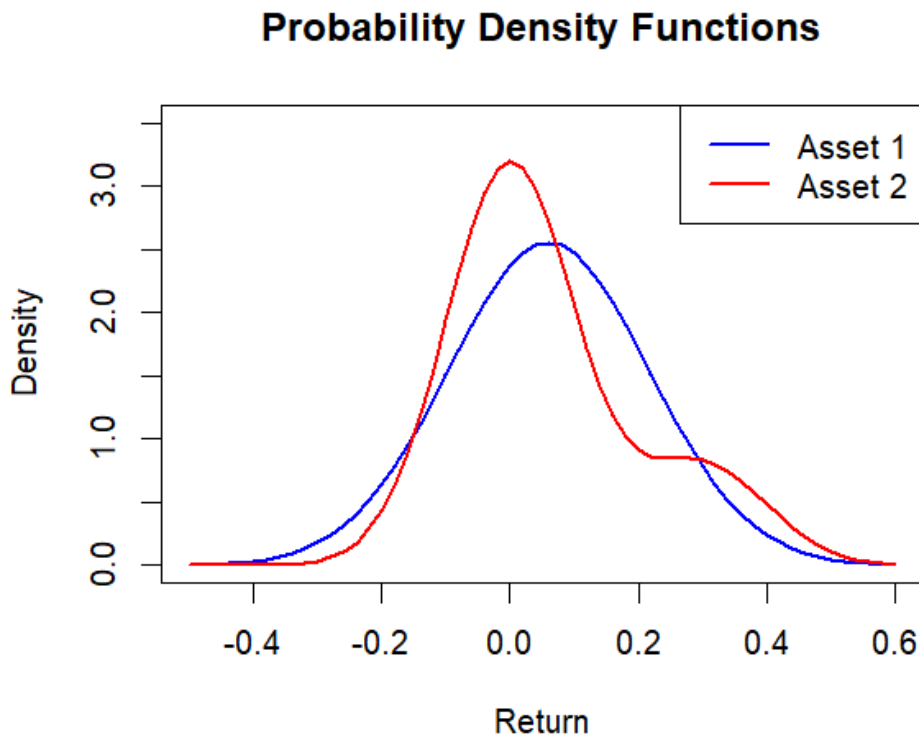


Figure 3: Probability density functions of returns for assets 1 and 2

(f) The cumulative distribution function plot for asset 1 shows a gradual increase and consistent (normal) whereas CDF of Asset 2 shows major peak near 0% (which has 80% probability state) and a minor peak near 30% (which has 20% probability state). The peaks for asset two are sharper (narrow spread) due to low variability in each state.

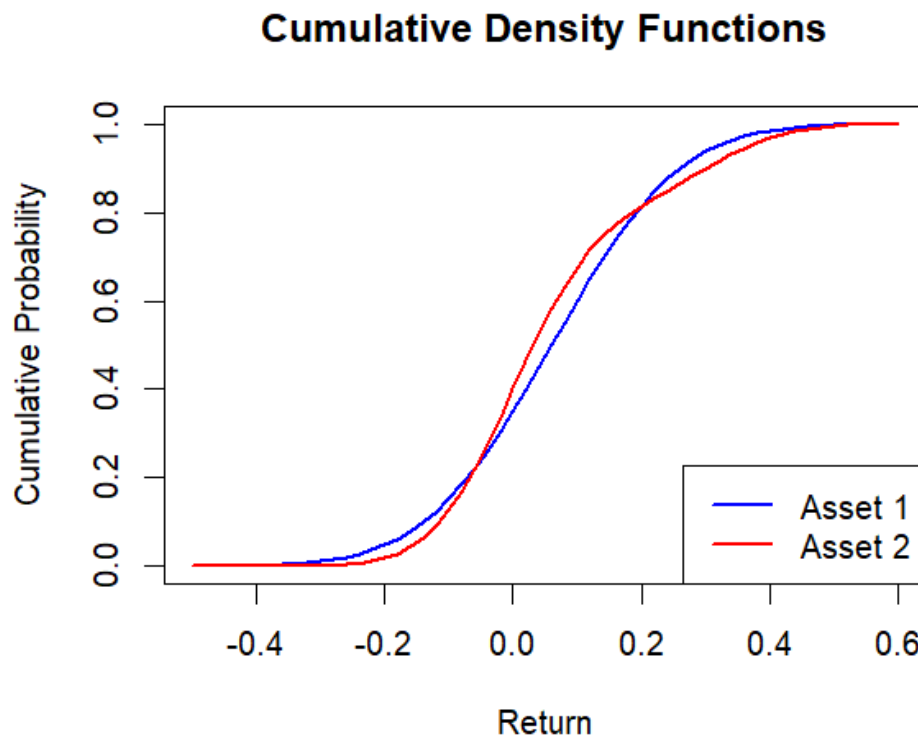


Figure 4: Cumulative density functions of returns for assets 1 and 2

(g) Asset 2 has a higher probability of small losses (40% vs. 35%) but a lower chance of extreme losses (12.7% vs. 15.3%) than Asset 1. Unlike Asset 1's unimodal normal distribution, Asset 2 is bimodal, making returns less predictable. It's density plot shows higher peak, but fatter tails, indicating volatility. Asset 1 has a smoother risk profile with fewer extreme movements. Asset 2 suits investors prioritising avoidance of large losses over small, frequent ones.